

What Feminist Theory & Praxis Says About Internet Networking



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(She/Her/Hers)

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Some Barn in Petaluma

Content Note

- “Feminism” is a contested term
- Talk mentions:
 - Stonewall Uprising
 - How sexism and racism can intersect
 - Poverty
 - Other oppressions mentioned quickly

Format

- Network Problem or Curiosity
- Feminist theory and/or praxis
- Network Solution/Explanation

“My Application is Slow”



“My Application is Slow”

```
for record in local_data:  
    post_to_remote_server(record)
```

Constraints

- 1,000,000 records, each 1KB in size
- 100 Gb/s connections between servers
- 50 ms latency (round trip) between servers

We run the app.

It takes 50,000 seconds.

50,000s = ~14 hours

Slack to network engineer:

We need more bandwidth between DFW & NYC.

So...

Network engineer says, “You have 100 Gbps!
It’s super fast! It’s not congested!”

Slack to network engineer: **It’s still slow!**

Let's do the math!

1,000,000 records x 1,024 bytes: ~1GB

1GB x 8 (to convert to bits) = 8Gb

8Gb / 100 Gbps = .08s = 80ms

80ms is roughly two movie frames.

Constraints

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“My Application is Slow”

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    post_to_remote_server(record)
```

Let's do the math!

$$1 \text{ second} / 50\text{ms} = 20$$

Only 20 sequential round trips can be done in one second!

What do Feminists Focus On?

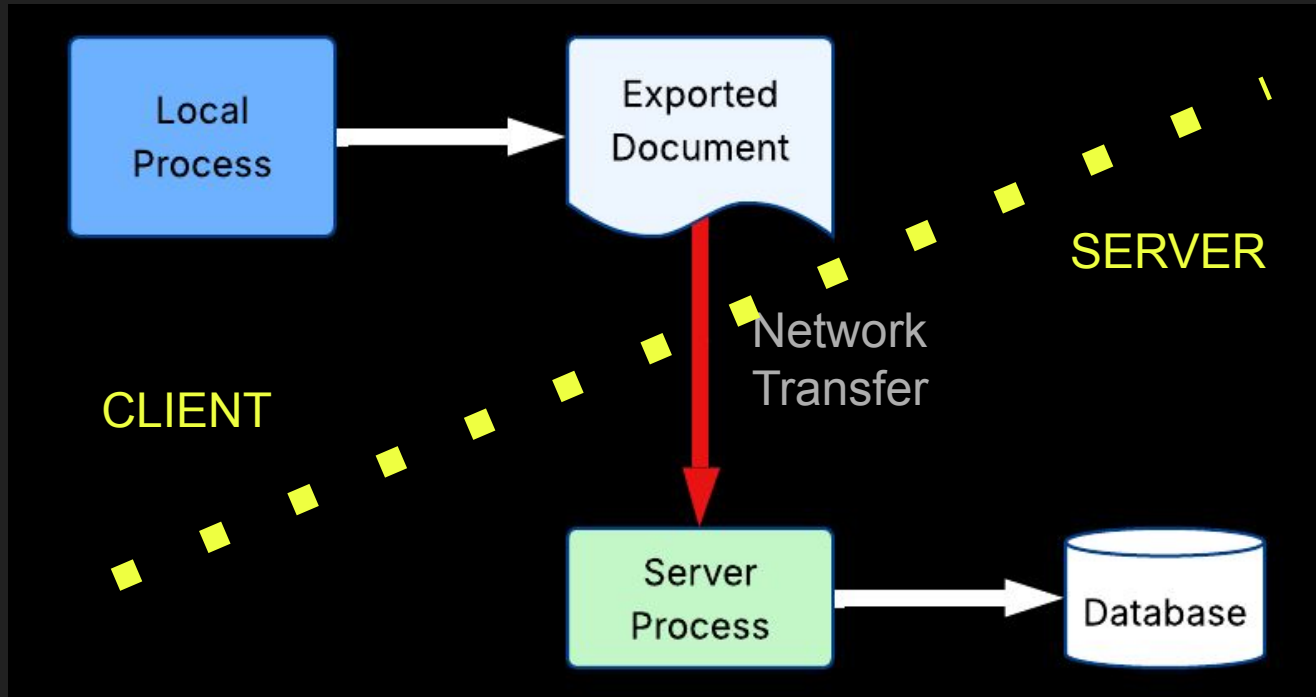
- Equal pay
- Transnational exploitation of feminized labor
- Reproductive justice
- Ending gender-related violence
- Reducing disparate impact of disease & disaster
- Ending war
- Lots of other stuff

A Better way

```
for record in local_data:  
    add_record_to_batch(record)  
  
send_batch_to_remote_server()
```

Freya Bhushan Mehta said this very eloquently in her talk yesterday, when she talked about the cost of crossing boundaries.

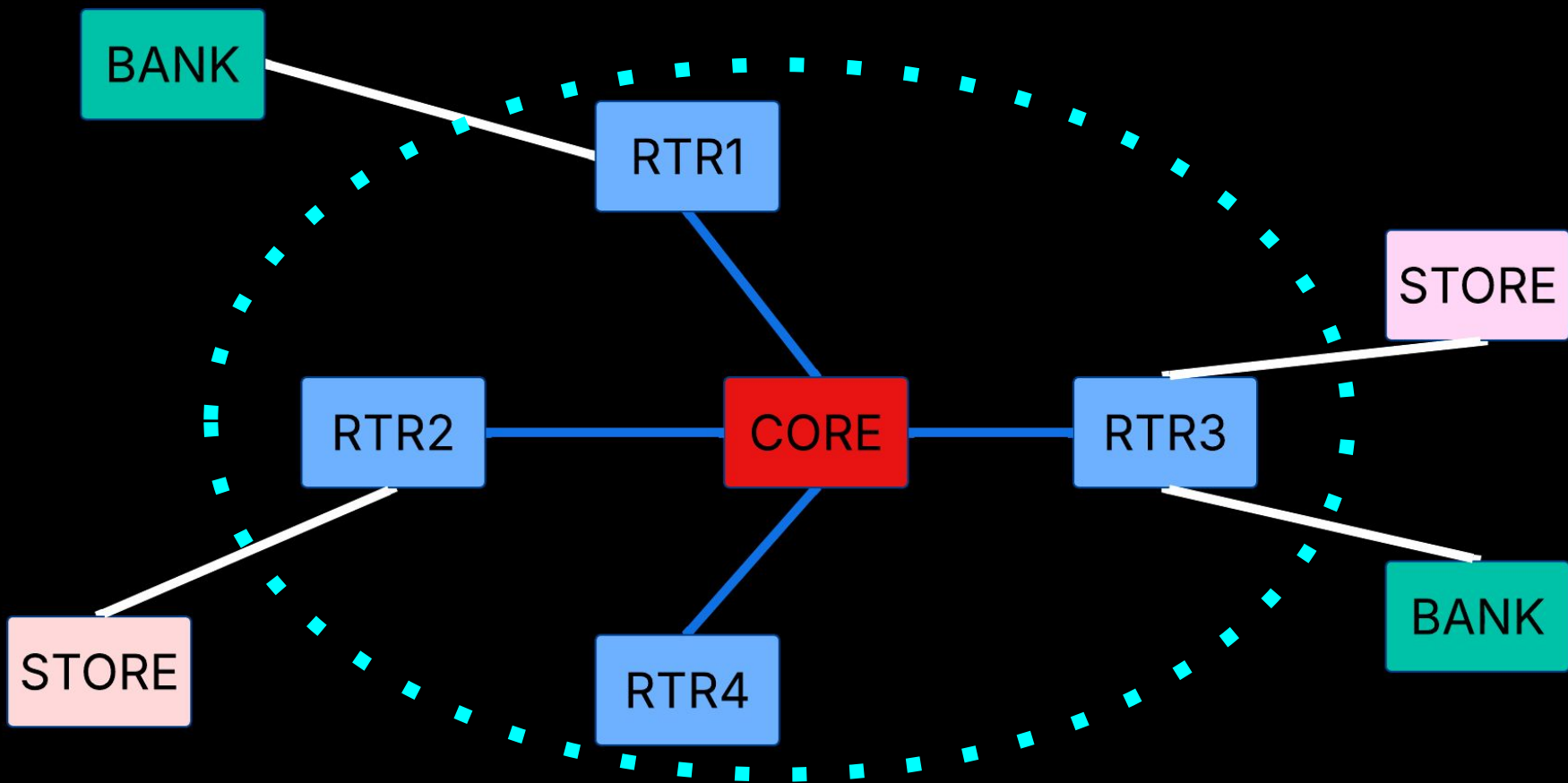
Let's do it a different way.



Takeaway

In Feminism and Networks, don't sequence tasks unnecessarily!



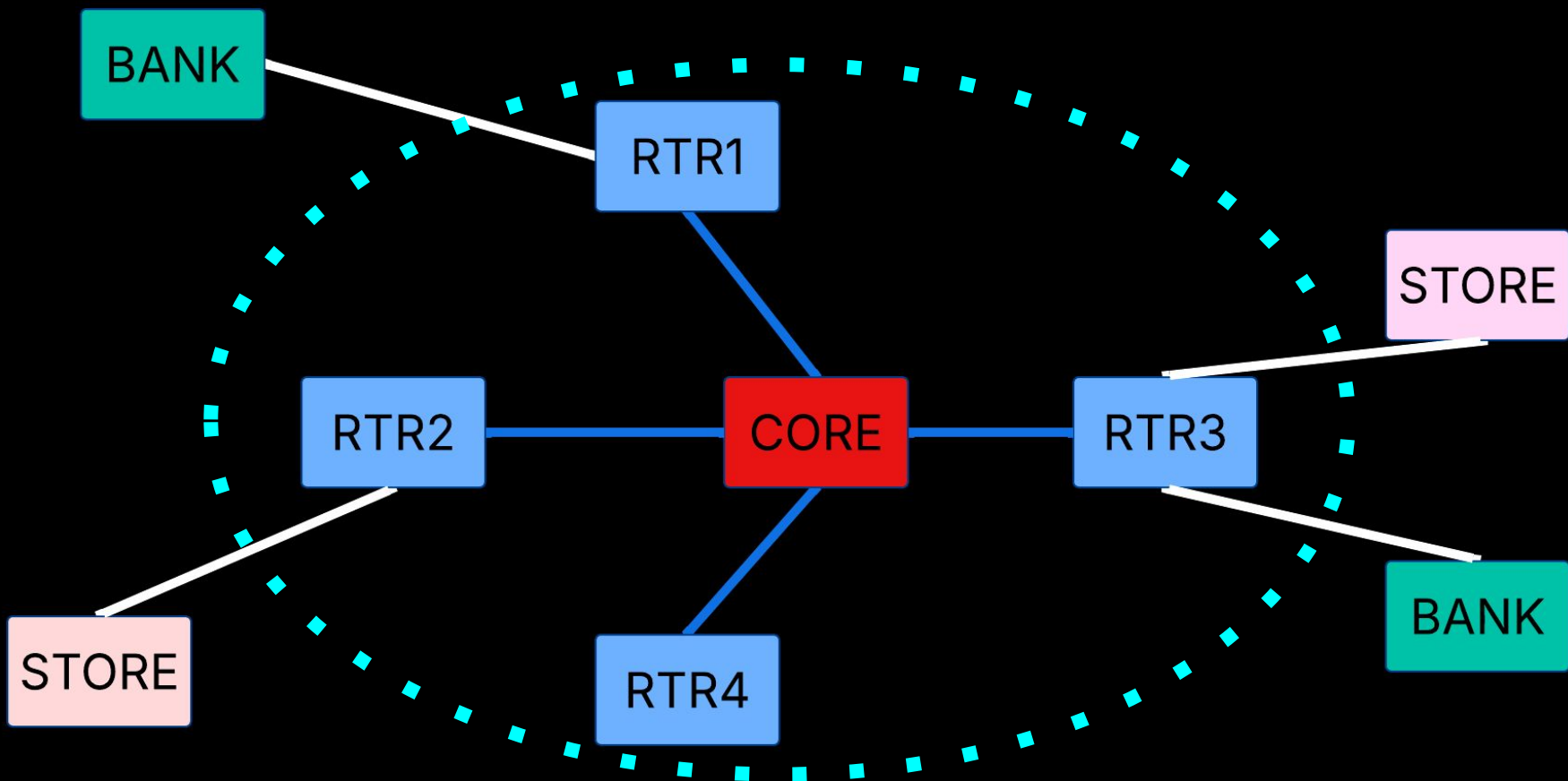


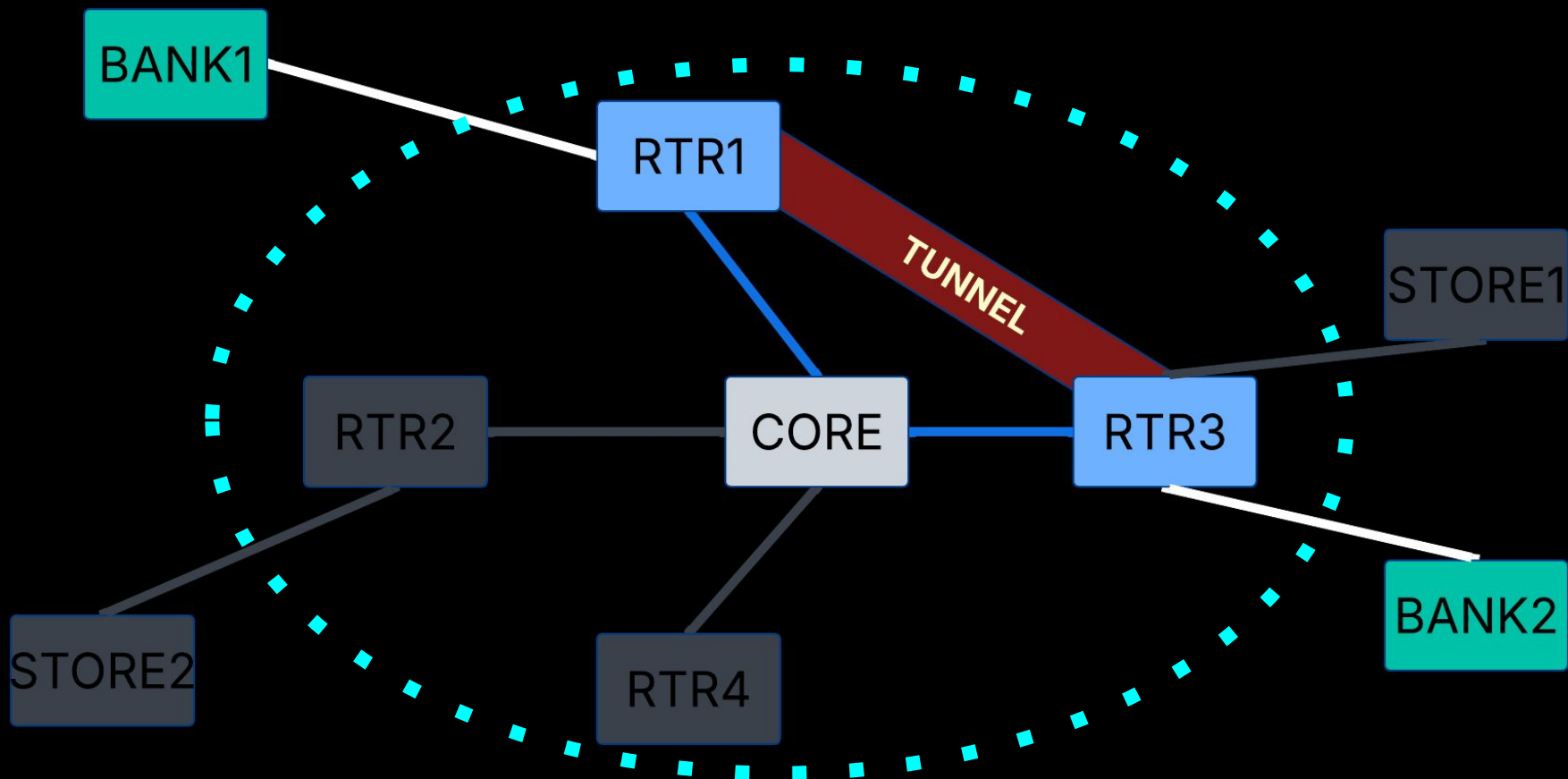
LGBT Rights

- Direct action (Stonewall Uprising, AIDS protests)
- Mutual aid
- Political lobbying
- Appealing to capitalism
- Family lobbying (PFLAG)
- Lawsuits
- NGOs

You can't do everything

- What are you good at?
- What energizes you rather than drains you?
- Each person only does *some* of the work



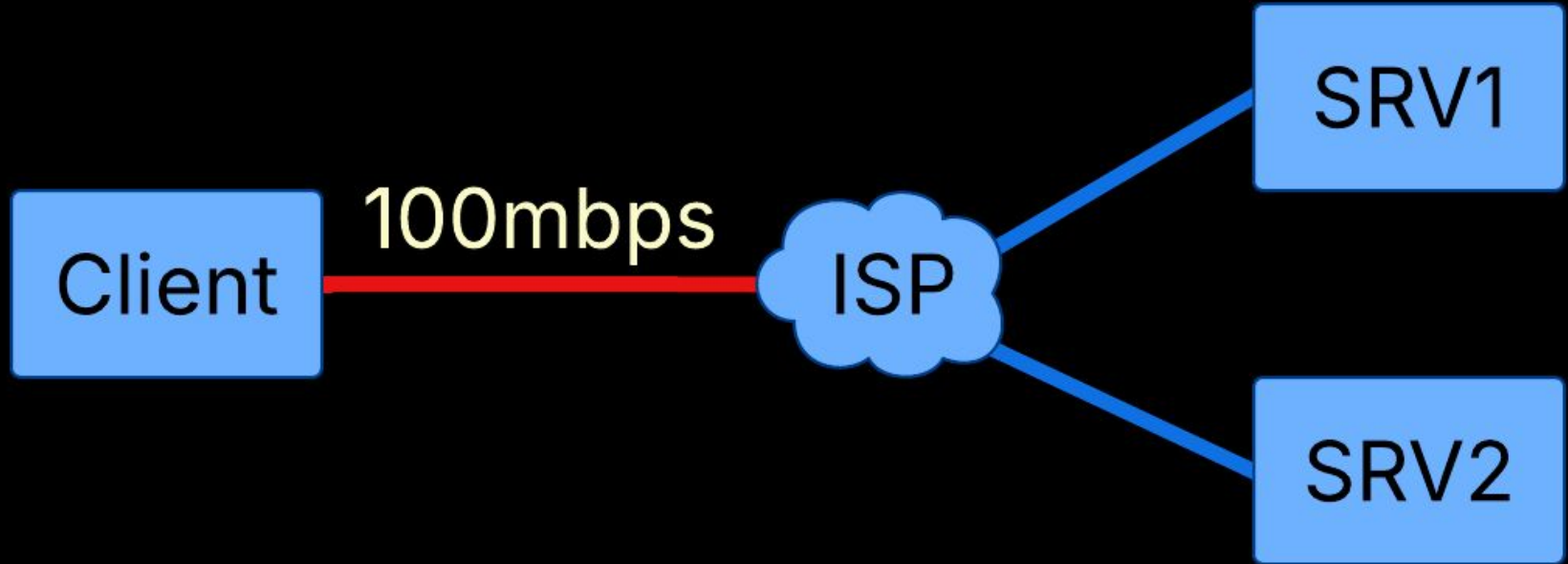


Takeaway

Do what is scalable/sustainable.

You can't do everything.

TCP “Fairness”



TCP “Fairness”

Real world simultaneous TCP sessions

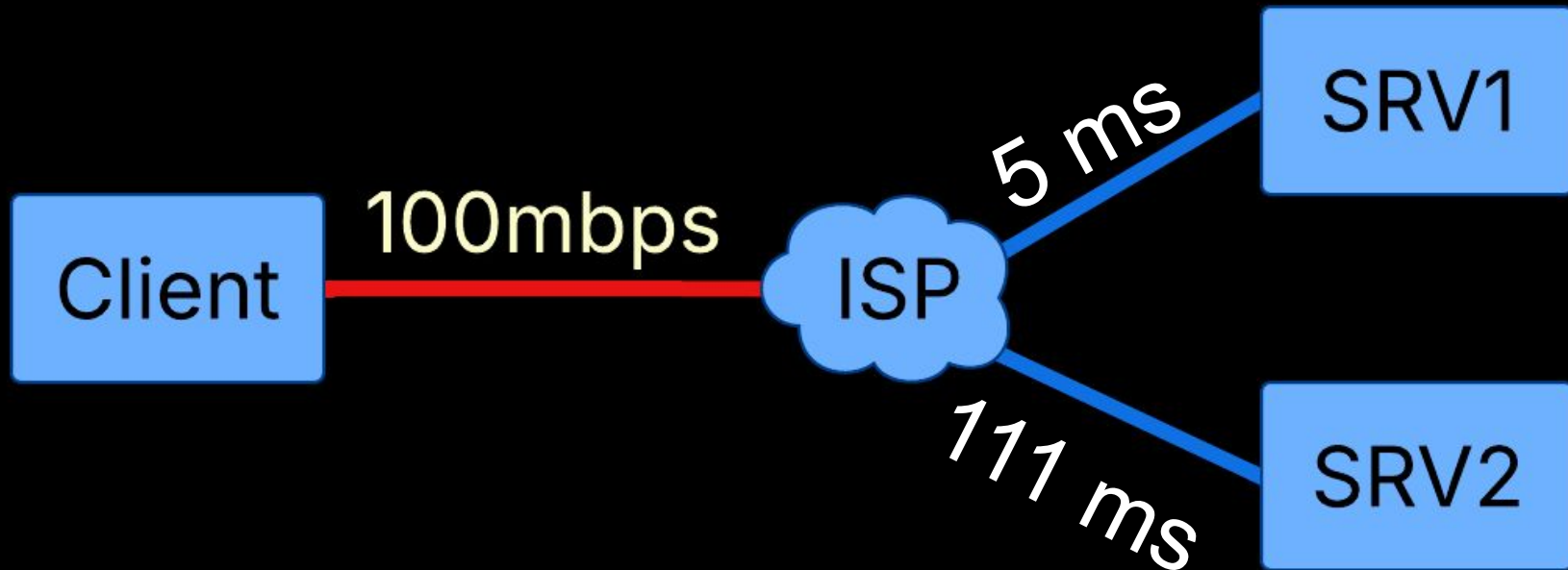
- Session A - 52.8 Mb/s
- Session B - 32.8 Mb/s

52.8 $\neq$ 32.8

Intersectionality

- Two oppressions combine to produce a new kind of oppression

TCP “Fairness”



TCP “Fairness”

Real world simultaneous TCP sessions

- Session A - 5ms RTT → 52.8 Mb/s
- Session B - 111ms RTT → 32.8 Mb/s

Takeaway

Two adverse conditions can combine to create a distinct adverse outcome

An MTR Issue

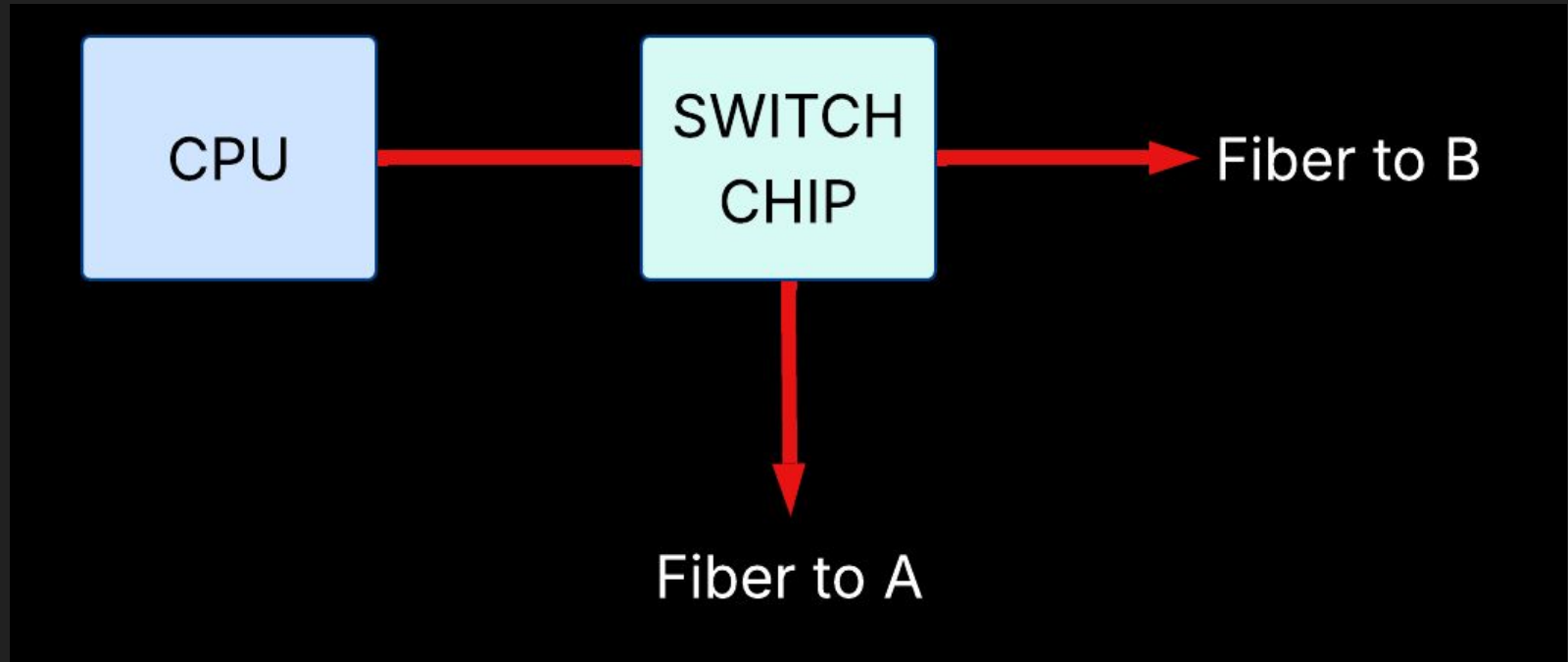
Host	Packets	
	Loss%	Snt
1. 192.168.157.1	2.6%	39
2. hlrn-dsl-gw03.hlrn.qwest.net	0.0%	38
3. 63-225-124-17.hlrn.qwest.net	0.0%	38
4. (waiting for reply)		
5. ae2.3607.ear5.denver1.net.lumen.tech	64.9%	38
6. 142.251.202.242	0.0%	38
7. 142.251.154.119	0.0%	38

Standpoint Theory

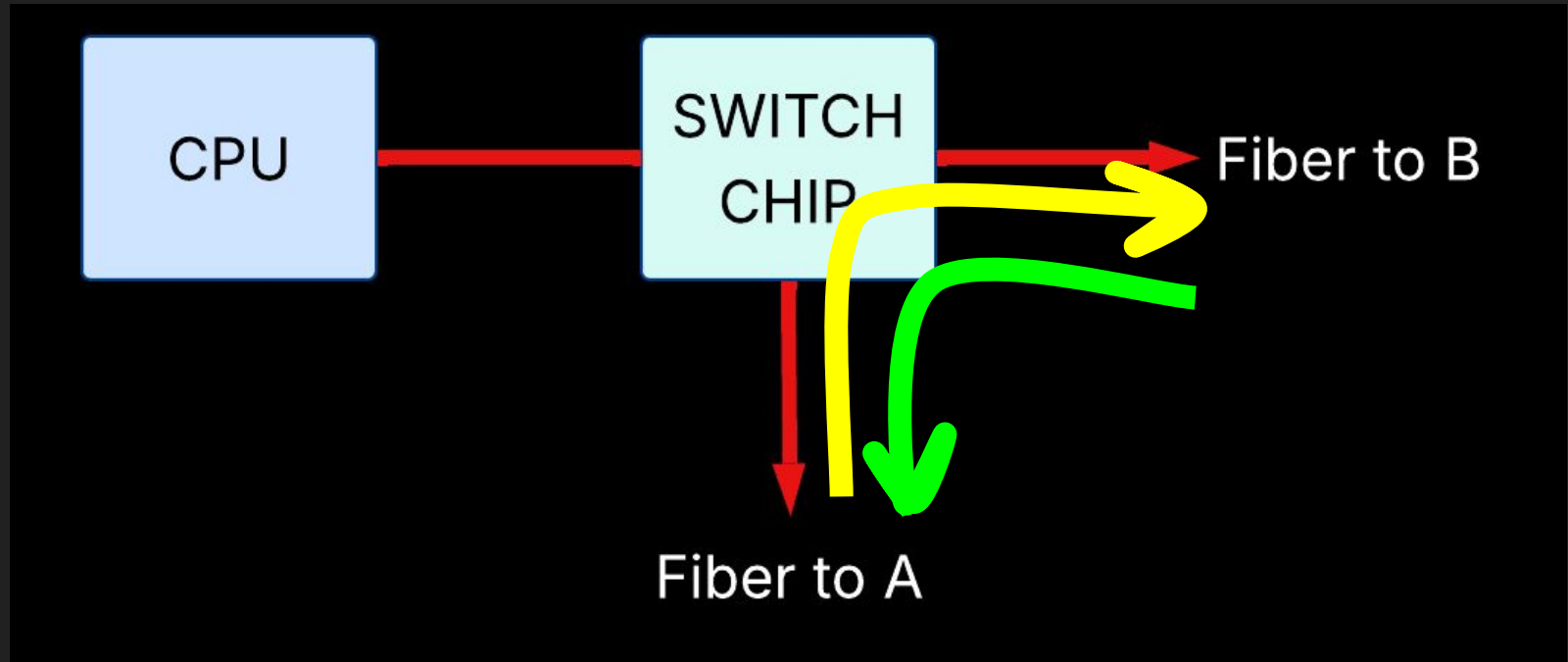
- The knowledge you can produce depends on where you are situated in the world
- Marginalized people have knowledge others do not have

(there's another part I didn't mention, find me if you want to know more!)

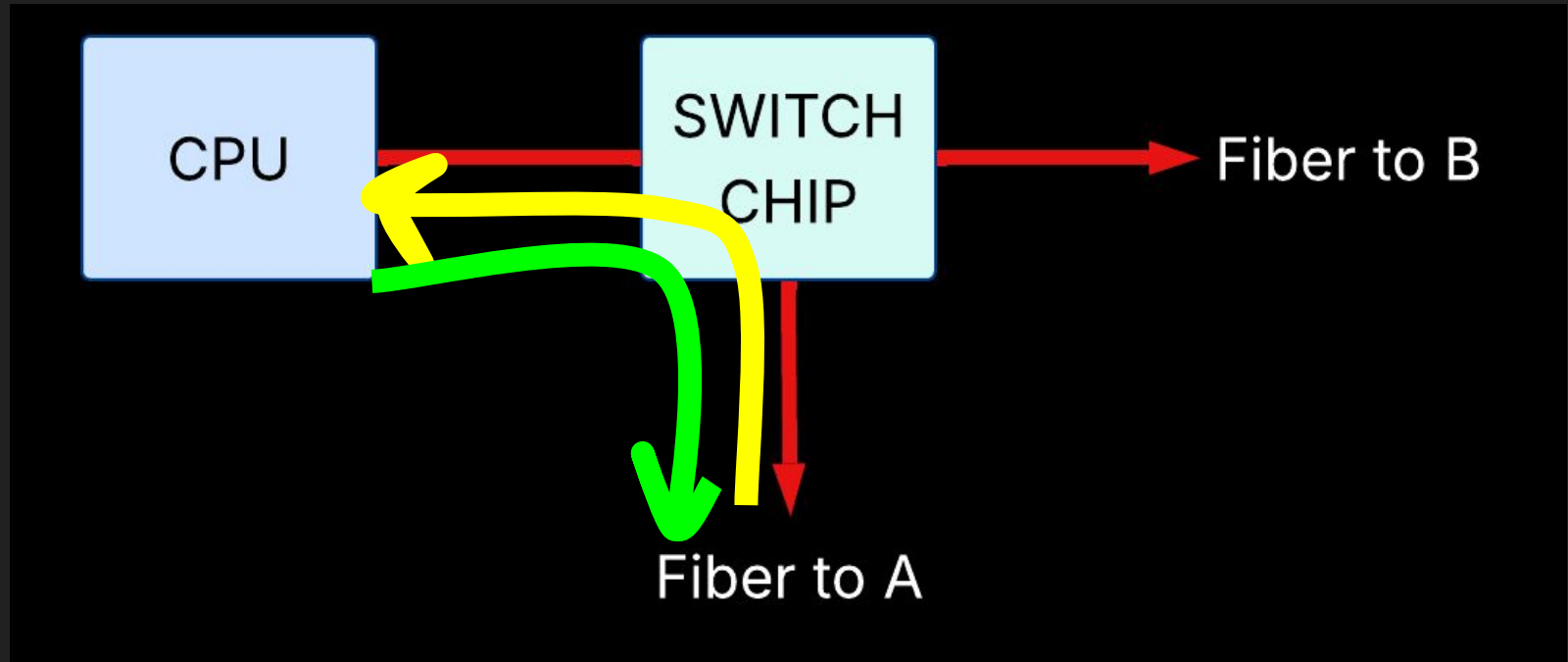
Router Architecture



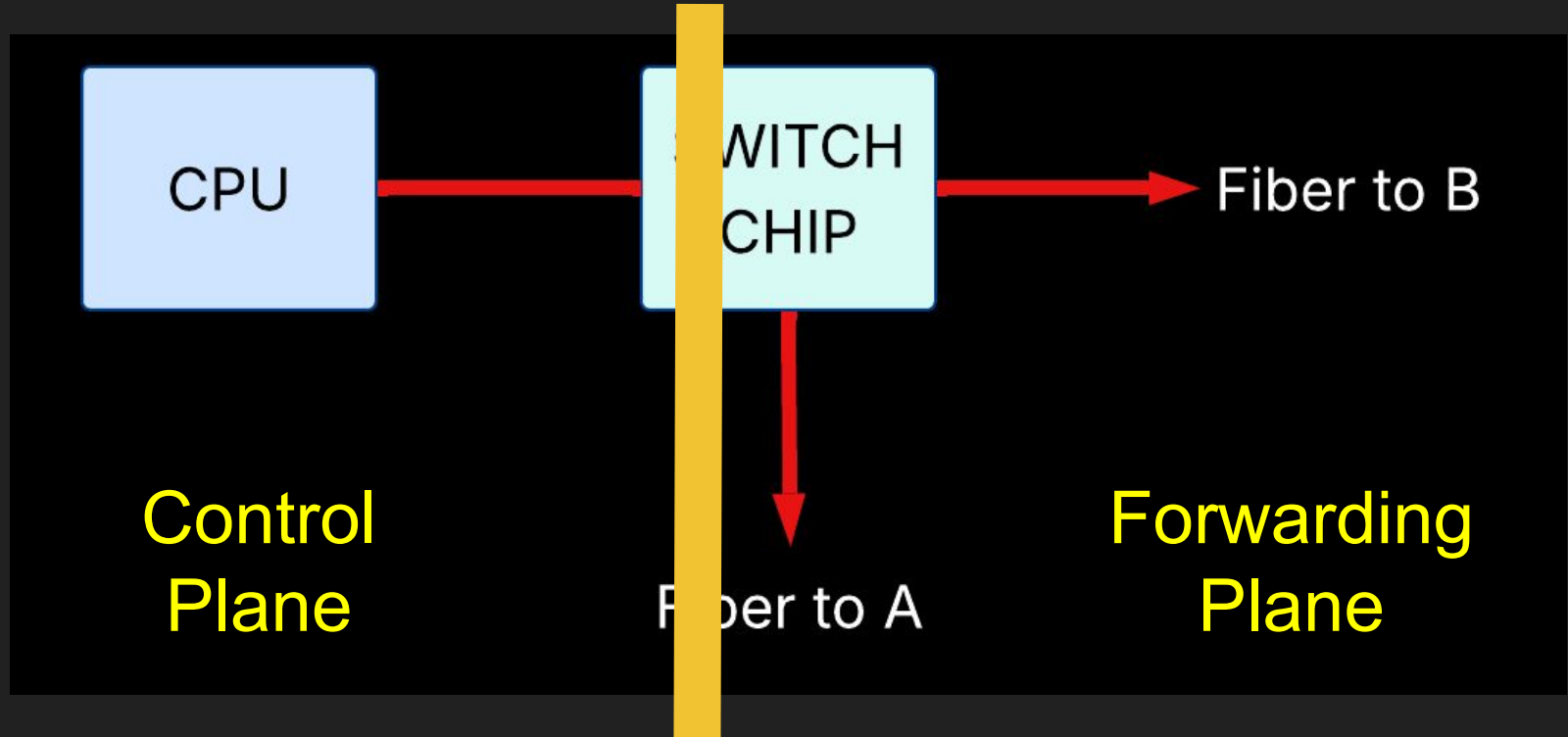
Router Architecture



Router Architecture



Router Architecture





Control Plane

**DATA
PLANE**

An MTR Issue

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Takeaway

Where someone (or something) is situated matters!

Understanding (Networks|People)

- Be curious!
- We can do more than one thing at once
- Scalability / sustainability is important
- Multiple difficulties can combine to create new ones
- Where something is situated matters

Where do you go from here?

- bell hooks
- Kimberlé Crenshaw
- Audre Lorde
- Julia Serano
- Combahee River Collective
- Patricia Hill Collins
- Leah Lakshmi Piepzna-Samarasinha
- Oh, also: NANOG, RIPE, APRICOT, AfNOG, etc.



Feminist Reading List

<https://tinyurl.com/bdz5934h>